# Anish Animesh Sahu

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## Summary

Computer Engineering undergraduate with hands-on experience in full-stack development, building web and mobile applications with React Native, React.js, Node.js, and Supabase. Skilled in developing AI-powered applications using C#, Python, OpenCV, and Tesseract OCR, with a background in computer vision and machine learning. Experienced in data analytics using Python (NumPy, Pandas, Matplotlib, Seaborn) and SQL, and comfortable designing clean, intuitive interfaces in Figma. Passionate about learning emerging technologies and applying them to solve real-world problems.

## Education

**K J Somaiya College of Engineering** — Mumbai
*B.Tech in Computer and Communication Engineering* — *2023 – 2027*

**Champions Academy** — Mumbai
*Higher Secondary Education* — *2021 – 2023*

**Atomic Energy Central School - 4** — Mumbai
*Secondary Education* — *2009 – 2021*

## Projects

**Civic AI – Mobile Civic Issue Reporting Platform**
*React Native, Supabase, Google Gemini, Expo*
- Developed a cross-platform civic reporting application using React Native and Supabase, enabling real-time issue reporting and tracking.
- Integrated Google Gemini for AI-based image analysis to classify issue category, urgency, and description automatically.
- Designed secure authentication, storage, and role-based access control using Supabase to ensure scalability.
- Built live map views and officer dashboards to improve issue visibility and response workflow efficiency.

**Housing Price Prediction Pipeline**
*Python, Scikit-Learn, Pandas*
- Built a predictive housing price model using Scikit-Learn by evaluating Random Forest, Decision Trees, and Linear Regression algorithms.
- Designed an automated data preprocessing pipeline for feature scaling, categorical encoding, and missing value imputation.
- Validated model performance using 10-fold Cross Validation and RMSE metrics to ensure strong generalization on unseen data.

**OCR-Based Text Recognition System**
*C#, .NET Windows Forms, OpenCvSharp, Tesseract OCR*
- Developed a desktop OCR application using C# Windows Forms and Tesseract OCR to detect and extract text from images.
- Implemented image preprocessing techniques using OpenCvSharp, including grayscale conversion, thresholding, and noise reduction, to improve text recognition accuracy.
- Designed a multi-region text detection pipeline with bounding box visualization for identifying and extracting text from individual regions within an image.
- Built an interactive GUI supporting image upload, OCR processing, and CSV export of extracted text for efficient data analysis.

## Experience

**Bhabha Atomic Research Centre (BARC)** — Mumbai
*Summer Intern – Text Recognition from Visual Images* — *May 21, 2026 – July 21, 2026*
- Developed a desktop application for text recognition from visual images using C#, .NET Framework, Windows Forms, OpenCvSharp, and Tesseract OCR.
- Implemented image preprocessing techniques including grayscale conversion, thresholding, and noise reduction to improve OCR accuracy on input images.
- Designed a multi-region text detection pipeline with bounding box visualization and region-wise text extraction for enhanced document analysis.
- Integrated CSV export functionality and optimized the OCR workflow through testing, debugging, and performance improvements.

## Technical Skills

**Languages:** C#, JavaScript, Python, SQL, C++, HTML/CSS

**Machine Learning:** Scikit-Learn, TensorFlow, Keras, PyTorch

**Computer Vision:** OpenCvSharp, Tesseract OCR

**Data Analytics:** NumPy, Pandas, Matplotlib, Seaborn

**Frontend:** React.js, React Native, Expo

**Backend:** Node.js, Express.js, MongoDB, .NET, Windows Forms, Supabase

**Tools & Platforms:** Git, GitHub, Visual Studio, VS Code, Figma